ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. AI-DS ENGINEERING — SEMESTER 6

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Artificial Neural Networks (ANN): Perceptron model, Activation functions (ReLU, Sigmoid, Tanh). [14 Marks]

[OR]

Q.1 (Alt) Derive Backpropagation algorithm using Chain Rule for multi-layer feedforward networks. [14 Marks]

--- UNIT II ---

Q.2 Explain vanishing and exploding gradient problems. Discuss remedies: Batch Normalization, He/Xavier initialization. [14 Marks]

[OR]

Q.2 (Alt) Describe Optimizers: SGD with Momentum, RMSProp, and Adam optimizer with adaptive learning rates. [14 Marks]

--- UNIT III ---

Q.3 Explain Convolutional Neural Networks (CNN): Convolution layer, Stride, Padding, Pooling, and Flattening. [14 Marks]

[OR]

Q.3 (Alt) Describe classic CNN architectures: LeNet, AlexNet, VGG-16, and ResNet skip connections. [14 Marks]

--- UNIT IV ---

Q.4 Explain Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) cell architecture. [14 Marks]

[OR]

Q.4 (Alt) Describe Gated Recurrent Units (GRU) and Bidirectional RNNs for sequence modeling. [14 Marks]

--- UNIT V ---

Q.5 Explain Attention Mechanism and Transformer Architecture: Self-Attention and Multi-Head Attention. [14 Marks]

[OR]

Q.5 (Alt) Describe Generative Adversarial Networks (GAN): Generator and Discriminator minimax game. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****